

Pallavi Reddy Rangasamudram

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Available: From December 2026, full-time

PROFESSIONAL SUMMARY

Test and validation engineer completing an M.Eng. in Electrical and Microsystems Engineering at OTH Regensburg, with almost two years at an automotive access-systems supplier across cybersecurity, development tooling and test engineering. Master thesis: a robotic test bench for door-handle ECUs, from arm assembly and sensor integration to closed-loop CAN testing in Vector CANoe and probe testing from $-40\,^{\circ}\mathrm{C}$ to $+80\,^{\circ}\mathrm{C}$. Earlier: business automation software at Persistent Systems, India.

PROFESSIONAL EXPERIENCE

Minebea Access Solutions Deutschland GmbH — Erdweg, Germany

Master Thesis, Test Engineering – Robotic Test Automation for Door Handle ECUs — Oct 2025 – May 2026

Analysis and improvement of the Lock/Unlock test automation strategy with an ECU-controlled robotic arm

- Identified a finger-simulation probe that matched human capacitive response across $-40\,^{\circ}\mathrm{C}$ to $+80\,^{\circ}\mathrm{C}$, by testing six probe configurations at 1–5 N contact force, grounded and ungrounded, against measured human-finger reference values for an OEM door handle.
- Automated the capacitive Approach and Unlock tests with up to two autonomous retries, confirmed in an R&R study, by building a closed-loop CAN setup: C/C++ firmware on an Arduino GIGA (ARM Cortex-M4, TJA1051 transceiver) and CAPL evaluation of ECU signals in Vector CANoe.
- Brought a 3D-printed 6-axis robotic arm from kit to repeatable target-position replay (five axes integrated, four verified), by assembling the drive train, wiring stepper drivers, magnetic encoders and limit switches, and resolving gear binding by reprinting at adjusted scale.
- Defined an improved test strategy for push, pull, Lock and Unlock actions, after analysing the limits of the existing 3-axis tester (LabVIEW runtime, NI DAQ, PCAN, CAN/LIN/K-Line).
- Integrated NFC key-card, capacitive and strain gauge (force) sensors into the test environment; flashed OEM handle ECUs for each software release and extended the existing CAPL test library for the feedback loop.
- Alongside the thesis: ran functional validation tests with the validation team on door handles after climatic chamber exposure and at $60\,^{\circ}\mathrm{C}$, recording error values per handle and action; completed in-house soldering training.

Working Student – Electronics Development — Jan 2025 – Sep 2025

- Reduced manual planning work for project managers and team leads, by configuring and administering Codebeamer ALM and building tracker automation, a resource planning tracker for team availability, and time and leave planning templates.
- Took over Codebeamer administration and worked fully independently within a short time.

Student Intern / Project Thesis – Automotive Cybersecurity — Jul 2024 – Oct 2024

- Kept the Cybersecurity Case of an OEM project consistent with ISO/SAE 21434 and UNECE R155, by correcting the CAL field and its inheritance, extending system trackers with CAL derivation, restricting TARA status changes and restoring links to the standard in Codebeamer.
- Supported the S/MIME secure e-mail rollout for OEM project members (certificates, private keys), reported issues to the vendor and handled certificate requests and revocations via GLPI tickets.

Persistent Systems Ltd. — Pune, India

Associate Software Engineer (previously Student Intern) — Feb 2022 – Nov 2022

- Replaced manual registration and record-keeping steps for client business units, by developing business process automation applications and interfaces in Appian with SQL and Python.

Primark, Regensburg | *Sales Assistant (part-time), daily customer communication in German* — Jun 2026 – Present

EDUCATION

Ostbayerische Technische Hochschule (OTH) Regensburg — Regensburg, Germany

M.Eng. Electrical and Microsystems Engineering; preliminary GPA 2.5 — Oct 2023 – expected by Mar 2027

Project Thesis graded 1.3;
modules incl. Advanced Semiconductor Technology, Advanced Packaging, Advanced Optical Sensing

Sreyas Institute of Engineering and Technology (JNTUH) — Hyderabad, India

B.Tech Electronics & Communication; GPA 2.2, First Class with Distinction — Aug 2018 – Jul 2022

PROJECTS AND SEMINARS

- **Smart Car Parking Control System** (B.Tech project): Arduino-based gate control with entry and exit sensors, a servo-driven barrier and an LCD showing free spaces.
- **Cyber Threat Detection with Artificial Neural Networks** (B.Tech project): neural-network classification of security events based on event profiles.
- **Seminars:** Thin Gate Oxide Layers in Semiconductor Devices (M.Eng.); Hadoop Distributed Storage and Processing (B.Tech).

TECHNICAL SKILLS

- **Test & Validation (Prüfung, Validierung):** HIL test setups, Functional testing, Climatic and temperature testing, Test procedures, R&R studies, Test reports, Failure analysis
- **Vehicle Electronics:** CAN, Vector CANoe, CAPL, ECU flashing, NFC and capacitive access systems
- **Hands-on & Integration:** Robotic arm assembly and commissioning, Wiring, Sensor integration (capacitive, NFC, strain gauge, encoders), Stepper drivers, Soldering, 3D printing
- **Embedded & Programming:** C/C++, ARM Cortex-M4, Arduino, UART, Python (basic), SQL
- **Measurement:** Multimeter, Oscilloscope, Logic analyser
- **Tools & Processes:** Codebeamer ALM, Traceability, V-model, ASPICE, ISO/SAE 21434, UNECE R155, GLPI, Fusion 360 (model review), Microsoft Office
- **Familiar with:** LabVIEW / NI DAQ test systems, LIN, K-Line, ROS2 (self-study)

CERTIFICATIONS AND WORKSHOPS

- Secure Hardware Components in Automotive, Strategische Partnerschaft Sensorik e.V., Regensburg 2024
- Appian Certified Associate Developer 2022
- Microsoft Certified: Azure Fundamentals (AZ-900) 2022
- Workshops: PCB Designing (2019), Internet of Things (2019), Projects using Cisco Packet Tracer (2021)

LANGUAGES

English (C1, fluent); **German** (A2 completed, B1 course in progress, daily spoken use at work); **Telugu** (native)

INTERESTS AND VOLUNTEERING

- **Regensburg Women's Cricket:** team player in regional tournaments; volunteer organiser of cricket promotion sessions at OTH Regensburg.
- **Campus Radio:** hosted and scripted English programmes during the B.Tech.
- **Community Service:** organised learning and nutrition-awareness activities for children.